

# Yijun YUAN

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## Research Interests

- 📖  $p$ -adic Galois representations
- 📖  $p$ -adic transcendental number theory
- 📖 Formalization of arithmetic geometry

## Employment

2025–2027    📖 **Postdoctoral Fellow, Institute for Theoretical Sciences, Westlake University**  
Advisor: Huayi Chen

## Professional Position

2026–    📖 **Member, Committee on Mathematics and Artificial Intelligence, Chinese Mathematical Society**

## Education

- 2022–2025    📖 **Ph.D., Yau Mathematical Sciences Center, Tsinghua University**  
Pure Mathematics  
Thesis title: *The Field of Norms and  $(\varphi, \Gamma_{\mathfrak{F}, K})$ -Modules for  $p$ -Adic False Tate Curve Extensions.*  
Advisor: Lei Fu / Haoran Wang
- 2018 – 2021    📖 **M.Sc., Shanghai Center for Mathematical Sciences, Fudan University**  
Pure Mathematics  
Thesis title: *Uniformizer of the False Tate Curve Extension of  $\mathbb{Q}_p$ .*  
Advisor: Shanwen Wang / Quanshui Wu
- 2014 – 2018    📖 **B.S., School of Mathematical Sciences, East China Normal University**  
Mathematics and Applied Mathematics

## Research Publications

### Journal Articles

- 1    **Y. Yuan**, “Formalization of non-archimedean functional analysis 1: Spherically complete spaces,” *Journal of Symbolic Logic*, 2026, To appear.
- 2    S. Wang and **Y. Yuan**, “Hyper-algebraic invariants of  $p$ -adic algebraic numbers,” *Canadian Mathematical Bulletin*, 2026, To appear.  DOI: 10.4153/S0008439526102252
- 3    S. Wang and **Y. Yuan**, “Truncated expansion of  $\zeta_p^n$  in the  $p$ -adic mal’cev-neumann field,” *Chinese Annals of Mathematics, Series B*, 2026, To appear.

- 4 S. Wang and Y. Yuan, “Uniformizer of the false Tate curve extension of  $\mathbb{Q}_p$  (II),” *International Journal of Number Theory*, vol. 20, no. 01, pp. 1–18, 2024.  DOI: 10.1142/S1793042124500015
- 5 S. Wang and Y. Yuan, “Uniformizer of the false Tate curve extension of  $\mathbb{Q}_p$ ,” *The Ramanujan Journal*, vol. 58, no. 2, pp. 549–595, 2022.  DOI: 10.1007/s11139-021-00473-0

## Preprints (all submitted)

- 1 S. Wang and Y. Yuan, *p-adic hahn series with sparse support*, 2026. arXiv: 2607.06944 [math.NT].
- 2 Y. Yuan, *Descent of  $(\varphi, \tau)$ -modules in characteristic  $p$* , 2025. arXiv: 2511.17029 [math.NT].
- 3 S. Wang and Y. Yuan, *On the  $p$ -adic transcendence of  $\sum_{k=1}^{\infty} p^{-1/p^k}$* , 2025. arXiv: 2509.24609 [math.NT].
- 4 Y. Yuan, *Formalization of Harder-Narasimhan theory*, 2025. arXiv: 2509.19632 [math.AG].
- 5 Y. Yuan, *Multivariable period rings of  $p$ -adic false tate curve extension*, 2025. arXiv: 2505.24064 [math.NT].

## Books & Textbooks

- 1 H. Chen, J. Jiang, S. Wang, L. Xiao, Y. Yuan, and P. Yu, *Formalized Mathematics & Proof Assistant Lean*. 2026, Textbook in preparation.

## Talks

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|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026, Sept. 18th |  <b>Formalization of Non-Archimedean Functional Analysis</b><br><i>School of Mathematical Sciences, Fudan University</i>                                                                        |
| 2026, Jul. 20th  |  <b><math>p</math>-adic mathematics in Mathlib of Lean 4</b><br><i>Lean 4 Formalization Code Review Training Workshop</i><br><i>Peking University</i>                                           |
| 2026, Apr. 17th  |  <b>On the algebraicity of <math>p</math>-adic Hahn series</b><br><i>Jinling Conference on Number Theory and Arithmetic Geometry</i><br><i>Nanjing University</i>                               |
| 2026, Apr. 16th  |  <b>Formalization of Non-Archimedean Functional Analysis</b><br><i>Workshop on Formalized Mathematics</i><br><i>Yau Mathematical Sciences Center, Tsinghua University</i>                       |
| 2026, Apr. 9th   |  <b>Formalization of Non-Archimedean Functional Analysis</b><br><i>School of Mathematical Sciences, East China Normal University</i>                                                            |
| 2026, Apr. 7th   |  <b>Formalization of Non-Archimedean Functional Analysis</b><br><i>AI for Mathematics @ SIMIS</i><br><i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i>                   |
| 2025, Nov. 19th  |  <b>Formalization of Harder-Narasimhan Theorey</b><br><i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i>                                                                  |
| 2025, Jun. 13th  |  <b>Arithmetic of spherically complete valued fields in mixed characteristic</b><br><i>Symposium on Number Theory and Algebraic Geometry of Spring 2025</i><br><i>Capital Normal University</i> |

## Awards

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- 2024     Outstanding Teaching Assistant in Undergraduate Fundamental Courses, Tsinghua University (Second Class)
- 2023     Scholarship of Yau Mathematical Sciences Center (Second Class)  
          Tsinghua University Comprehensive Excellence Scholarship (Second Class)

## Peer Review

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- 2024     Journal of Number Theory

## Teaching Experience

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- 2026, Autumn     TA for **Foundation of Algebra and Analysis (I)** at Westlake University. AI<sub>4</sub>Math training for undergraduates.
- 2026, Spring     TA for **Foundation of Algebra and Analysis (II)** at Westlake University. Formalization (Lean 4) training for undergraduates.
- 2025, Autumn     TA for **Foundation of Algebra and Analysis (I)** at Westlake University. Formalization (Lean 4) training for undergraduates.
- 2024, Autumn     TA for an undergraduate course on **Linear Algebra** at Tsinghua University.
- 2023, Autumn     TA for an undergraduate course on **Linear Algebra** at Tsinghua University.
- 2022, Autumn     TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

## Skills

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- Coding     Lean 4, Python (SageMath),  $\text{\LaTeX}$ , Mathematica, ...